Sanika Pawar

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SUMMARY OF QUALIFICATIONS

Android Developer and Computer Engineering diploma student at Government Polytechnic Pune (94.60% in 5th semester, 96.20% in 10th), specializing in Android app development, offline communication systems, machine learning (ML Kit) and cloud automation. Proficient in Python, Java, C, C++, JavaScript, HTML5, CSS3 and Android Studio, with a proven record of designing, building and publishing working applications from idea to release, including one live on the Google Play Store.

Published **WakeLock**, an Android alarm app on the Play Store that keeps ringing until the user completes a task, built in Java with ML Kit pose detection. Designed **AcousticSync**, an offline messaging app that sends data between phones over near-ultrasonic sound with a custom CRC-16 protocol, ACK/retry reliability and collision avoidance, as a capstone project in a 4-member team. Built a serverless, event-driven image-processing pipeline on GitHub Actions for cloud computing, a Lost and Found application in Java, and a cybersecurity case study on the Oldsmar water plant attack. Holds Infosys certifications in C++, Python and AI Fundamentals.

PROJECT EXPERIENCE

WakeLock – Task-Based Alarm App **Google Play Store**

Android Developer: Designed, built and published WakeLock, an Android alarm app that will not stop ringing until the user completes a given task, so oversleeping is no longer one tap away. Developed in Java using Android Studio, and learned console-based Java and Google AI/ML Kit pose detection (machine learning) to verify task completion through the camera. Owned the full product cycle: development, testing, Play Store listing and release.

[Achievements]

- Built, shipped and published a live Android app on the Google Play Store.
- Integrated AI/ML Kit pose detection into an Android application.

AcousticSync: Air-Gapped Data Transfer Tool **Capstone Project, Govt. Polytechnic Pune** **2026 – 2027**

Team Member (4-member team, faculty-guided): Built an offline peer-to-peer messaging app that lets nearby smartphones exchange short text messages using sound — no Wi-Fi, Bluetooth, mobile data or central server — for disasters, outages and remote areas. Data travels through the speaker and microphone as near-ultrasonic signals (about 15–20 kHz) encoded and decoded with the ggwave library. Designed a custom application-layer protocol (TYPE:SENDER:SEQ:PAYLOAD:CRC) with CRC-16 error detection, sequence numbers, ACK-based retransmission (up to 4 attempts) and duplicate suppression, plus CSMA/CA-style listen-before-talk with random backoff and startup noise-floor calibration.

[Key Features]

- Peer discovery with HELLO/PING frames and a live acoustic radar; SOS emergency broadcast with repeated transmission and a banner alert on receiving devices.
- Message relaying to extend range; real-time spectrogram and signal/noise monitoring on HTML5 Canvas using the Web Audio API (AnalyserNode).
- Offline persistence with IndexedDB for messages, peers and SOS logs; ultrasonic/audible mode toggle for devices with weak high-frequency hardware.
- Tested across two Android/browser devices: permissions, CRC rejection, duplicates, SOS and persistence. Roadmap: AES-GCM end-to-end encryption, TTL-based multi-hop routing, unicast addressing.
- Technologies: JavaScript (ES6), HTML5, CSS3, Web Audio API, ggwave (WebAssembly), Capacitor, Android, IndexedDB, HTML5 Canvas.

Serverless Event-Driven Image Processing & Metadata Pipeline **Cloud Computing Project**

Developer: Built a serverless pipeline on GitHub Actions that runs with no manual steps: push an image to the *uploads/* folder and GitHub triggers a temporary cloud runner that sets up Python and Pillow, reads the image metadata (name, size, format), adds a red "CONFIDENTIAL – DEMO" watermark, saves it to *processed/*, commits the result back and shuts down. Demonstrates serverless, event-driven and pipeline concepts; chose GitHub Actions over GCP to get the same architecture without card or prepayment verification.

Lost and Found Application **Java Project**

Developer: Built a Lost and Found application in Java to report lost items and find them again.

Case Study | Oldsmar Water Plant Cyberattack **Cybersecurity**

Studied the 2021 Oldsmar incident, a remote attack on a city water plant, to understand how critical infrastructure is attacked and how it can be secured.

EDUCATION

Diploma in Computer Engineering, Government Polytechnic Pune, India – 5th Semester: 94.60%

Secondary School Certificate (10th), 96.20%

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, JavaScript, HTML5, CSS3

Android & AI: Android Studio, Android Development, Google ML Kit (Pose Detection), Machine Learning

Cloud & Tools: GitHub, GitHub Actions, Serverless Architecture, Event-Driven Automation, Capacitor

Concepts: Networking Protocols, Error Detection (CRC-16), Web Audio API, IndexedDB, Cybersecurity Fundamentals, Computer Science Fundamentals

CERTIFICATIONS, LANGUAGES & EXTRACURRICULARS

Certifications: C++ – Infosys | Python – Infosys | AI Fundamentals – Infosys | Yoga Certification (Yog Pravesh)

Languages: Hindi, English, Marathi | Learning Japanese

Extracurriculars: Calisthenics | Yoga